PYTHON

- What is Python?
- The Birth of Python
- History of Python
- Features of Python
- Versions of Python

The Python Environment:

- Installation of Python
- Python Documentation
- Getting Help
- Python Editors and Ides
- Basic Syntax
- Running a Python Script
- Python Scripts on UNIX/Windows

Getting Started:

- Keywords
- Data Types
- Variables
- Naming Conventions
- Print( )
- Type( )
- Id( ) Functions
- Input( )
- Raw_input() functions
- Conversion functions
- Del Keyword

String Handling:

- What is String?
- Single-quoted string literals
- Triple-quoted string literals
- String Indexing
- String Slicing

- Working with String Functions
- Working with String Methods

Operators:

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Assignment Operators
- Short Hand Assignment Operators
- Bitwise Operators
- Membership Operators
- Identity Operators

Flow Control:

- About Flow Control
- Conditional Statements (Simple if, if…else, if…elif…)
- Looping Statements (while loop, while … else, for
- loop, infinite loop, nested loops)
- Break
- Continue
- Pass

Sequences:

- About Sequences
- Lists
- Tuples
- Indexing and Slicing
- Iterating through a sequence
- Functions for all sequences
- Operators and keywords for sequences
- The xrange( ) function
- Nested sequences
- List comprehensions

Dictionaries and Sets:

- About Dictionaries
- When to use dictionaries
- Creating Dictionaries

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- Getting dictionary values
- Iterating through a dictionary
- Reading file data into a dictionary
- counting with dictionaries
- about sets
- Creating sets
- working with sets
- Working with Directories

Functions:

- Defining a function
- Calling a function
- Function Parameters
- Returning Values
- Variables scope
- Call by value
- Call by reference
- Passing collections to a function
- Passing functions to function
- Lamda function

Modules:

- What is a module?
- Creating user defined module
- Setting path
- The import statement
- Module search path
- From … Import
- Module Aliases
- Dir function
- Working with Standard modules
- (Math, Random, Date and time,Os and sys, String,….)

Errors and Exception Handling:

- Syntax Errors
- What is Exception?
- Need of Exception handling
- Predefined Exceptions

- Predefined Exceptions Hierarchy
- Except
- try
- finally clause
- Handling Multiple Exceptions
- User defined Exceptions
- Raise
- Assert statements

File Handling:

- Opening a file
- Closing a file
- Writing data to files
- Reading a data from files
- Tell()
- Seek() functions

OOPs concepts:

- About OO programming
- Defining Classes
- Creating object
- Class methods and data
- Static methods
- Private methods
- Constructors
- Destructors
- Inheritance
- Types of inheritances
- Polymorphism (over loading & over riding)
- Data hiding

Regular Expressions:

- What is regular expression?
- Wild card characters
- Forming regular expressions
- Character classes
- Quantifiers
- Greedy matches
- Grouping

- Match
- Search functions
- matching v/s searching
- Sub function splitting a string
- Replacing text
- Flags

Database Access:

- Basics of database
- Connections
- Executing queries
- Transaction management

Advanced concepts in python:

- Iterators
- Generators
- closure
- decorators
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Advanced Modules:

- Math sand sciences (Numpy, scipy, matplotlib)
- Data analysis (pandas)